36103 Statistical Thinking for Data Science

Assessment Task 3: Data analysis project for marketing campaigns

Calling the right customers: predicting subscription to a new telecommunication plan

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Date of submission: 1 November 2026

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Word count: 1,493 (main text, excluding title page, tables, figures, captions, references and appendices)

Code: QuangTrungNguyen_Assessment3_Code_01112026.ipynb (submitted with this report)

Executive summary

The company called 41,168 customers about a new plan, and 11.3% subscribed. Four models were tested on 8,234 unseen customers. Previous subscribers are the best prospects: 65% subscribed again, and they supply 19% of all subscribers. Customers aged 65 and over (47%) and under 25 (24%) respond far better than those aged 35 to 54 (9%), and mobile calls work better than landline calls.

Calling the top 20% of customers ranked by the logistic regression model would reach 65% of subscribers with 80% fewer calls. I recommend this model, a limit of three call attempts, and safeguards for older customers.

1. Problem formulation

The campaign made 36,531 calls that ended without a sale. The marketing manager and campaign planners need a ranked call list and simple rules, the analytics team maintains the model, and customers are the people it affects. The analysis shows which customers are worth calling and how far call timing explains the result.

RQ1 asks which segments are most likely to subscribe after adjusting for contact method, campaign history and economic conditions. RQ2 asks how accurately subscription can be predicted before a call, and whether a non-parametric model beats a parametric one. RQ3 asks which strategies follow; Table 1 lists the hypotheses.

Table 1. Hypotheses and statistical tests

RQ	Hypothesis	Test
RQ1	H1: Customers aged under 25 and 65+ have higher odds of subscribing than those aged 35-44, after adjustment	Wald z-tests; likelihood-ratio test (LRT) for the age block
RQ1	H2: A successful previous campaign is the strongest customer-level predictor	Adjusted odds ratios; permutation importance
RQ1	H3: Cellular contact raises the odds, and the channel effect differs by previous outcome	Wald z-test; LRT for contact × poutcome
RQ3	H4: More call attempts in the same campaign lower the odds	Wald z-test for log2(campaign)
RQ2	H5: A non-parametric model ranks customers better than logistic regression	Paired t-test on CV folds; bootstrap CI on test AUC; phase-3 test

2. Data and preparation

Table 2 lists data issues and treatments; three changes respond to the AT1 feedback. I kept all ages, because the 10 customers over 90 subscribed at 50%. Formal tests show that every predictor except `housing` and `loan` is associated with subscription ($p < 0.05$; Appendix A).

The rows are in time order: the `month` column wraps from December back to March twice, giving 26 month blocks in three campaign phases (Figure 1). Phase 1 averaged 3,954 calls a month at 4.8% success; phase 3 averaged 228 at 52.1%. Volume and success rate are negatively correlated across blocks (Spearman $\rho = -0.67$, $p < 0.001$), so AT1's strong months were late, low-volume months.

Table 2. Data issues and treatment (41,180 raw records, 41,168 after cleaning)

Issue	Evidence	Treatment
File format	Each row wrapped in quotes with doubled inner quotes	Repaired before parsing (as in AT1)
Dictionary mismatch	`balance` and `day` listed but absent; `day_of_week` present; `pdays` uses 999, not −1	Used the actual columns; documented
Duplicates	12 identical rows	Removed
Age 17-98	10 records over 90, 50% subscribed	Kept (AT1 removed them)
Target leakage	`duration` is known only after the call; CV AUC 0.93 with it vs 0.79 without	Excluded
`pdays`	96.3% coded 999; 4,108 rows have `previous` > 0 but `pdays` = 999	Dropped; `poutcome` used instead
`default`	3 'yes' (all non-subscribers), 8,595 'unknown'	Binary flag `default_unk`

Issue	Evidence	Treatment
housing, loan	Same 990 'unknown' rows; no association (p = 0.06 and 0.58)	Excluded
Small category	education = illiterate, n = 18	Merged with basic.4y for modelling
Economic indicators	VIF up to 64 (euribor3m, emp.var.rate, nr.employed)	Kept nr.employed, cons.price.idx, cons.conf.idx (VIF ≤ 1.4)
Class imbalance	11.3% subscribed	Stratified splits, PR-AUC, tuned threshold; no resampling
'unknown' values	Informative: default unknown subscribes at 5.2% vs 12.9%	Kept as a separate level

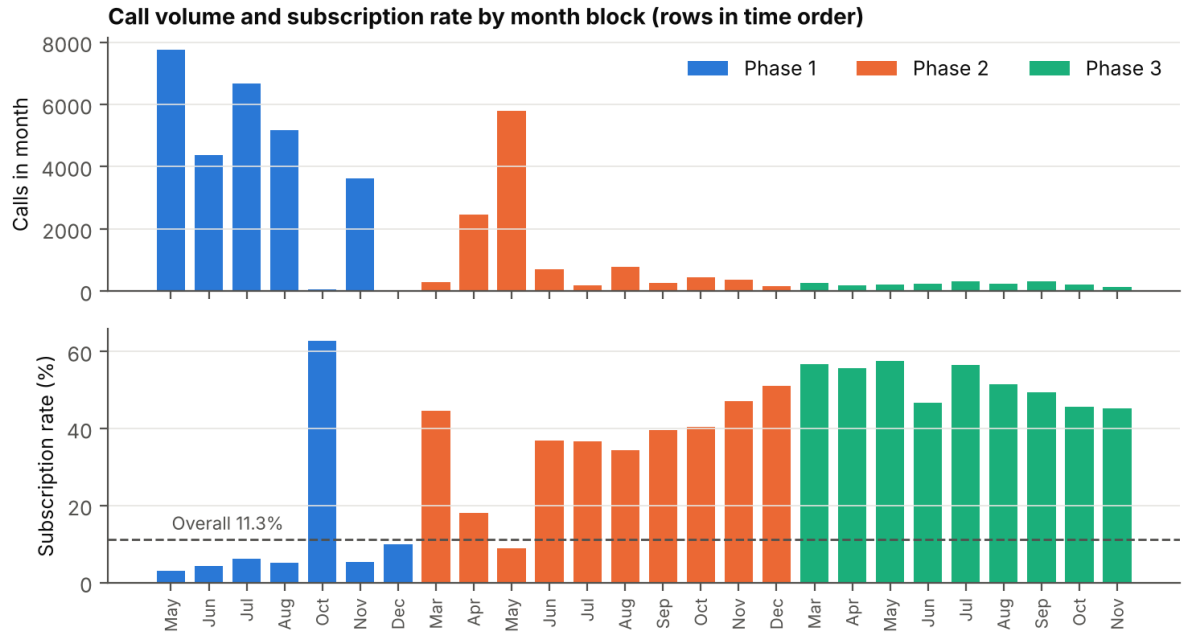


Figure 1. Calls and subscription rate for the 26 month blocks, coloured by campaign phase

3. Methodology

I split the data 80/20 into training (n = 32,934) and test (n = 8,234) sets, stratified by the target, and used only the training set for selection and tuning.

The parametric model is a logistic regression whose coefficients maximise the log-likelihood (maximum likelihood, Lecture 5). Its negative log-likelihood is convex, so the estimate is the global optimum. I added predictors in four blocks and compared nested models with LRTs, AIC and BIC (Appendix B). Age enters as six groups because a quadratic term confirmed a U-shape (LRT $p < 0.001$). The largest variance inflation factor is 3.3.

The non-parametric models use the same predictors, so accuracy differences come from model form. Each was tuned by grid search with 5-fold stratified cross-validation (CV) on ROC-AUC, because the business task is ranking.

The second Module 3 method is Bayesian estimation: the posterior is proportional to the likelihood times the prior. Segment rates use a Beta prior and Binomial likelihood. I also refitted M4 with Normal(0, 2.5^2) priors and sampled its posterior with Metropolis-Hastings MCMC, checking trace plots, autocorrelation, R-hat and effective sample size (ESS; Gelman et al., 2013). Table 3 gives each

model's settings. A 1,000-resample bootstrap tests the AUC difference, and a temporal test trains on phases 1 and 2 and predicts phase 3.

Table 3. Model choice, configuration and training

Model	Type and reason	Configuration and tuning grid	Selected
Logistic regression	Parametric. Binary outcome; coefficients give odds ratios	MLE (statsmodels). Blocks: profile, contact history, economy, timing. `nr.employed` chosen over `euribor3m` (AIC 18,484 vs 18,504)	Model M4, 47 parameters
KNN	Non-parametric. No functional form; similar customers behave alike	One-hot and standardised inputs. k: 100, 200, 400, 800, 1,600; uniform or distance weights	k = 800, uniform
Decision tree	Non-parametric. Readable segment rules	Depth: 4, 6, 8, 10. Minimum leaf: 50, 100, 200, 400	Depth 6, leaf 100
Gradient boosting	Non-parametric. Learns interactions and curves	Learning rate 0.05, 0.1; depth 2, 3, 5; leaf 100, 300, 1,000; L2 1, 10, 30; early stopping after 30 rounds (max 1,000 trees)	Rate 0.1, depth 3, leaf 300, L2 30, 535 trees. All 54 settings within 0.005 CV AUC
Beta-Binomial (Bayesian)	Module 3 estimation. Uncertainty for small segments	Prior Beta(2.25, 17.75): mean 11.3%, weight of 20 customers. Sensitivity prior Beta(1, 1)	Posterior mean and 95% credible interval
Bayesian logistic regression	Module 3 estimation. Posterior for the 47 coefficients of M4; no closed form, so MCMC	Normal(0, 2.5^2) priors on standardised predictors. Random-walk Metropolis-Hastings, 4 chains × 60,000 iterations, burn-in 5,000, thin 50. Sensitivity prior Normal(0, 0.5^2)	Largest R-hat 1.003; smallest ESS 1,199 of 4,400 draws

Note. All classifiers: 5-fold stratified CV on the training set, ROC-AUC scoring, F1-optimal threshold from out-of-fold predictions, random seed 42.

4. Results

4.1 Model comparison

All four models rank customers well (Table 4, Figure 2): test ROC-AUC is 0.797 to 0.807, and PR-AUC is about four times the no-skill value of 0.113. Gradient boosting leads logistic regression in CV by 0.008 AUC (paired t-test p = 0.013), but the 95% bootstrap CI for the test difference includes zero (−0.0003 to 0.015). The top 20% ranked by either model hold 65% to 66% of subscribers (Figure 3).

On phase 3, where `nr.employed` falls below every training value, AUC drops to 0.60 to 0.70. Models using only customer and contact variables reach 0.70, so customer patterns transfer and economic patterns do not.

Table 4. Model performance (CV on training set; test set n = 8,234; phase-3 test n = 2,054)

Model	CV ROC-AUC (sd)	CV PR-AUC (sd)	Test ROC-AUC	Test PR-AUC	Precision	Recall	F1	Top-20% capture	Phase-3 ROC-AUC
Logistic regression	0.784 (0.011)	0.439 (0.017)	0.800	0.462	0.448	0.586	0.508	0.648	0.674
KNN	0.783 (0.008)	0.407 (0.018)	0.799	0.434	0.450	0.577	0.506	0.654	0.602
Decision tree	0.780 (0.010)	0.427 (0.021)	0.797	0.450	0.464	0.586	0.518	0.636	0.697
Gradient boosting	0.792 (0.008)	0.455 (0.022)	0.807	0.477	0.472	0.591	0.525	0.660	0.647

Note. Phase-3 models were trained on phases 1 and 2. Customer and contact variables only: logistic regression 0.697, gradient boosting 0.698.

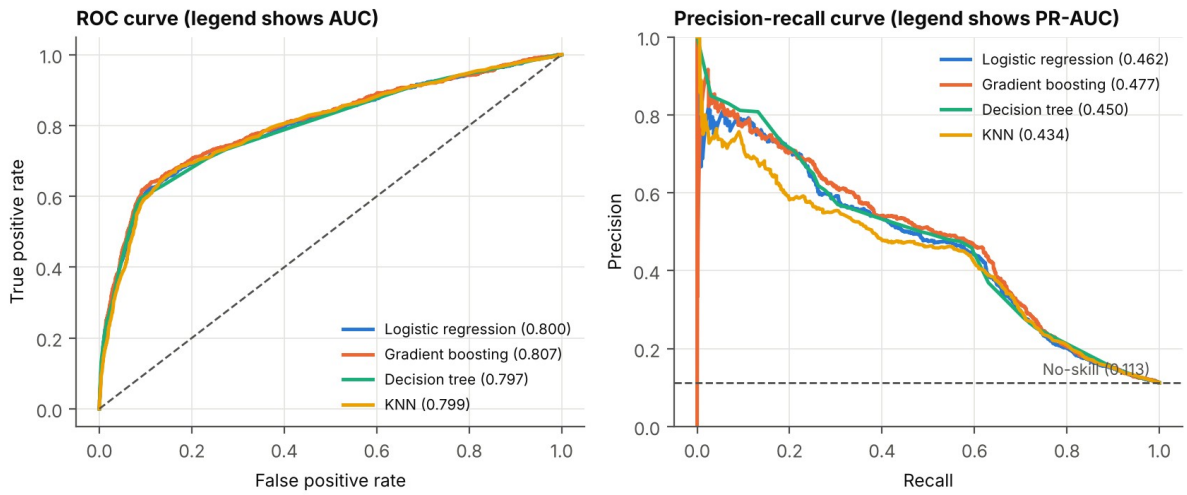


Figure 2. ROC and precision-recall curves on the test set

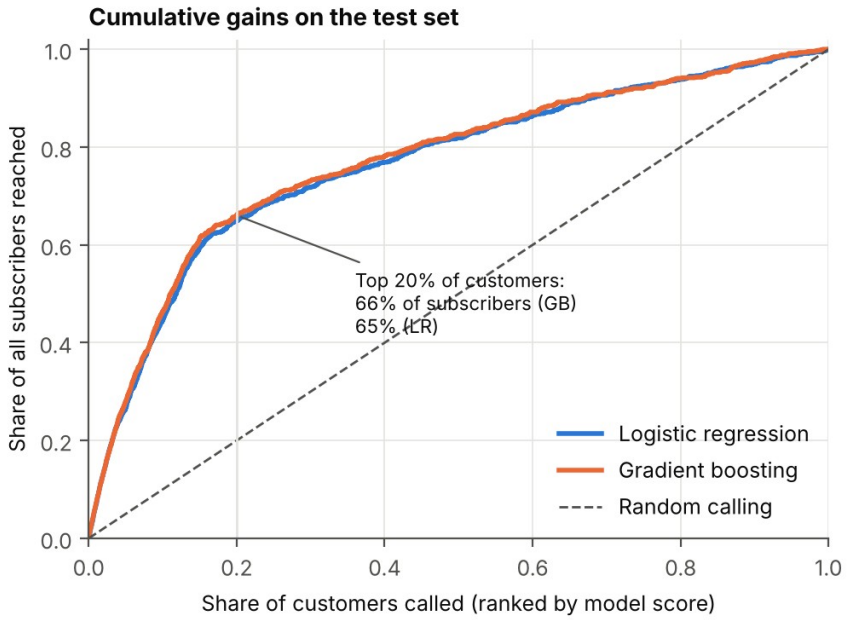


Figure 3. Cumulative gains: share of subscribers reached by calling the top-ranked customers

4.2 Parametric estimates

Figure 4 shows adjusted odds ratios (OR) from the final model, M4. A previous success raises the odds 3.68 times (95% CI 3.17 to 4.29). A previous failure lowers them (OR 0.63) although its raw rate is higher (14.2% vs 8.8%), because 84% of previous failures were called in phases 2 and 3. Telephone contact cuts the odds by 38% (OR 0.62), and each doubling of call attempts cuts them by 7% (OR 0.93).

Compared with ages 35-44, the odds are 1.48 times higher at 65+ and 1.30 times higher at 17-24. With age in the model, the student and retired effects are not significant (p = 0.25 and 0.18), so the AT1 job finding is mainly an age effect. Neither interaction (contact × previous outcome, contact × age group) improved the model (LRT p = 0.94, 0.20).

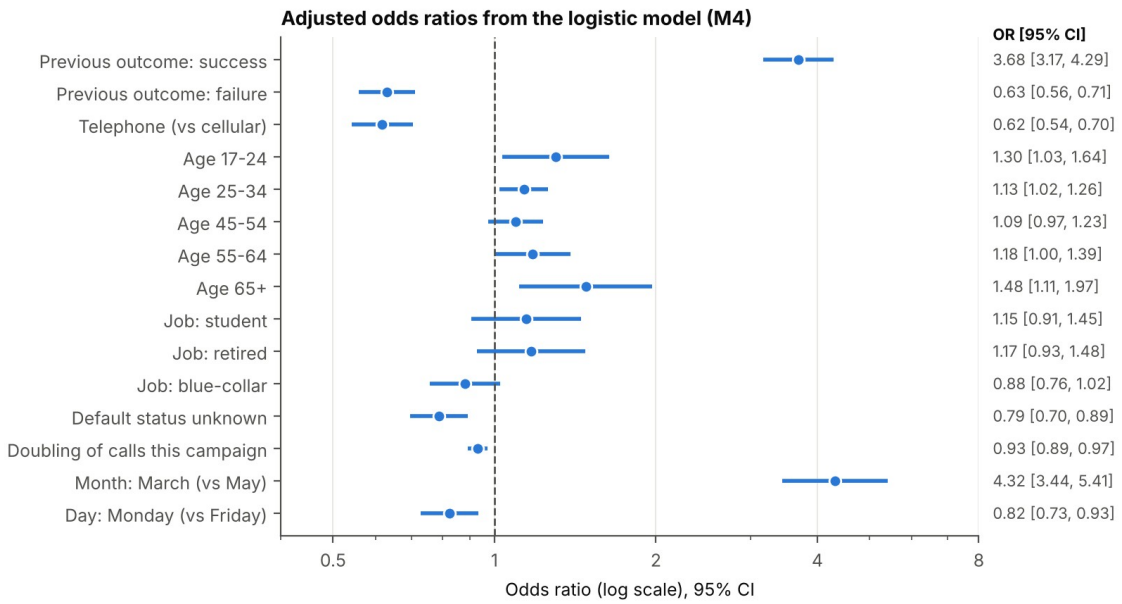


Figure 4. Selected adjusted odds ratios from logistic model M4 (reference levels: age 35-44, admin., cellular, never contacted, May, Friday)

4.3 Non-parametric findings

Permutation importance ranks `nr.employed` far above every customer variable (Figure 5). Age shows the same U-shape as in the logistic model, and the best tree segment is previous subscribers called when `nr.employed` was below 5,088 (64% to 77%; Appendix C).

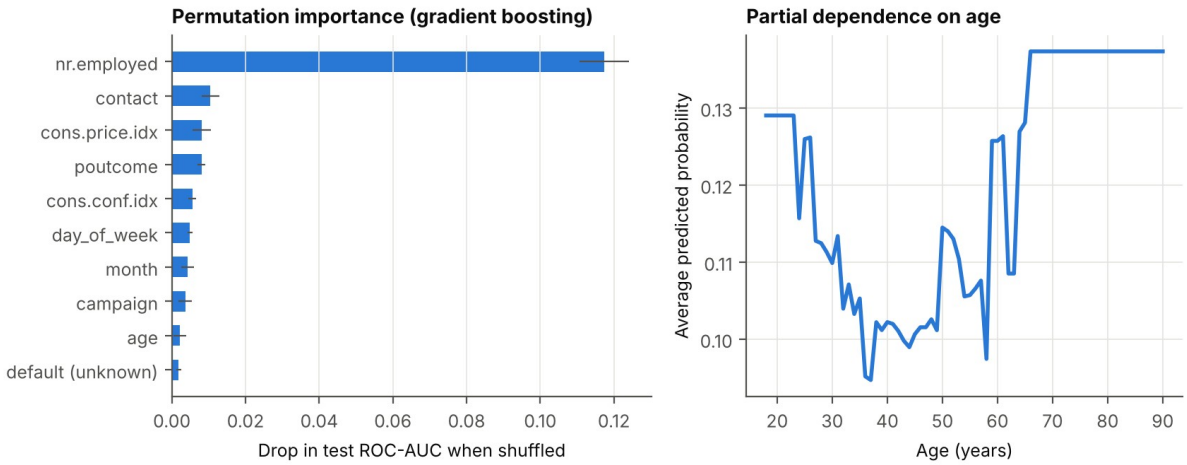


Figure 5. Gradient boosting: permutation importance on the test set (left) and partial dependence on age (right)

4.4 Bayesian estimation

Customers aged 65+ have a 95% credible interval of 42.6% to 50.0% for their rate (Figure 6). For the 18 illiterate customers, the prior moves the observed 22.2% to 16.5% with a wide interval (6.6% to 29.6%).

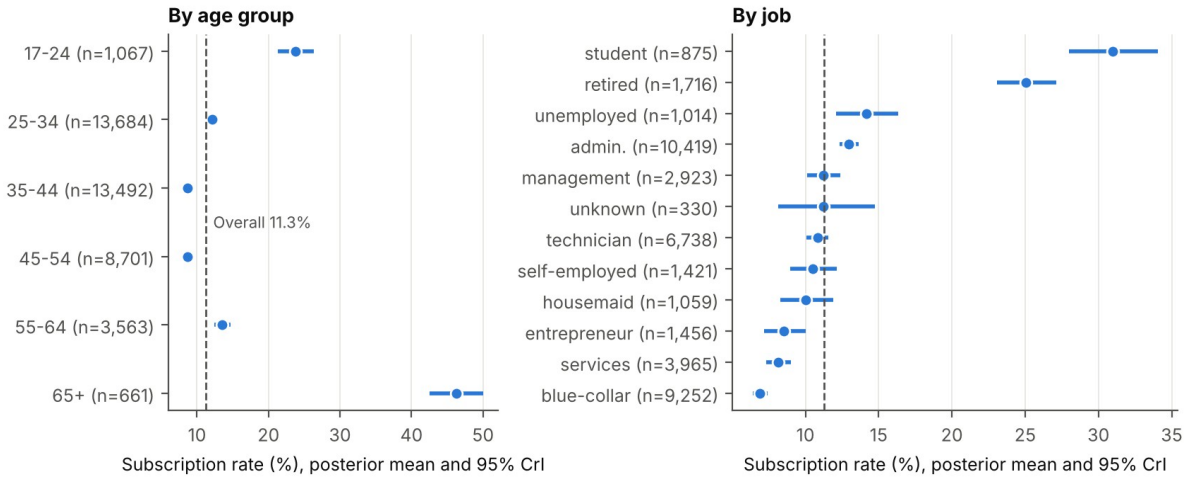


Figure 6. Posterior subscription rate by age group and job (Beta-Binomial, 95% credible intervals)

The Bayesian logistic regression converged (largest R-hat 1.003, smallest ESS 1,199; Figure 7). Its posterior odds ratios match the maximum likelihood estimates within 0.11 standard errors, so the data outweigh the prior, and a tighter prior moves no coefficient by more than 1.1 (Appendix G). Unlike a confidence interval, a credible interval gives probabilities directly: the odds ratio for a previous success exceeds 3 with probability 0.996.

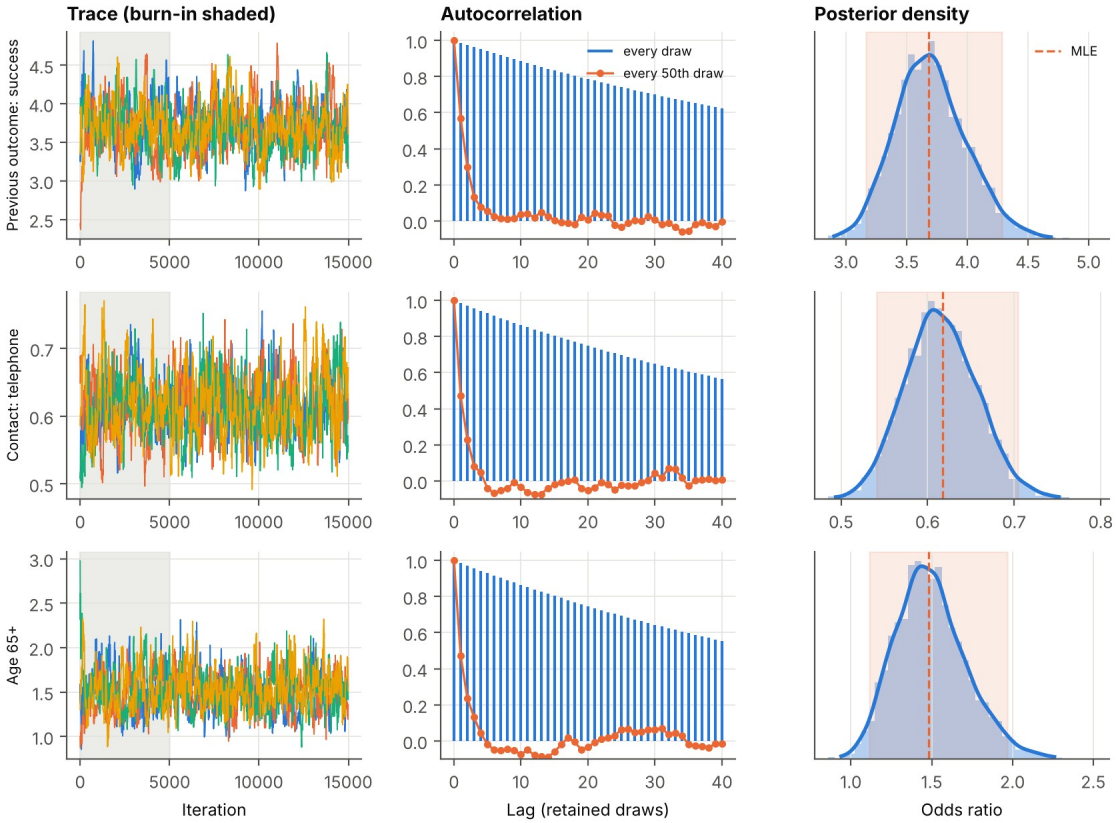


Figure 7. MCMC diagnostics for three odds ratios: trace of four chains (burn-in shaded), autocorrelation of every draw and of every 50th draw, and posterior density (dashed line and band: maximum likelihood estimate and 95% CI)

Table 5. Hypothesis outcomes

Hypothesis	Evidence	Decision
H1 age	17-24: OR 1.30 [1.03, 1.64]; 65+: OR 1.48 [1.11, 1.97]; age block LRT p = 0.018	Supported
H2 previous success	OR 3.68 [3.17, 4.29], the largest customer-level effect; `nr.employed` dominates overall importance	Supported at customer level
H3 channel	Telephone OR 0.62 [0.54, 0.70]; interaction LRT p = 0.94	Main effect supported; interaction not
H4 call attempts	OR 0.93 [0.89, 0.97] per doubling	Supported
H5 model type	CV gain +0.008 (p = 0.013); test CI includes 0; phase-3 AUC 0.647 vs 0.674	Not supported in practice

5. Business insights and strategy

Call previous subscribers first: they are 3.3% of customers but 19.2% of subscribers (Table 6). Design offers by age band, since job titles add little once age is known. Call mobiles and stop after three attempts: customers needing more are 18.5% of those called but 12.0% of subscribers.

I recommend the logistic model for deployment, refitted before each campaign. Its test AUC is within 0.007 of gradient boosting, it held up better in phase 3 (0.674 vs 0.647), and its coefficients can be audited.

Table 6. Insights, evidence and recommended actions

Insight	Evidence	Action	Measure
1. Previous subscribers convert best	65.1% rate; 3.3% of customers, 19.2% of subscribers; OR 3.68	Call this list first in every campaign	Conversion of the re-contact list
2. Age drives response more than job	65+: 47%, OR 1.48; 17-24: 24%, OR 1.30; job effects not significant after age	Design offers and scripts by age band	Conversion and complaints by age band
3. Mobile beats landline; extra attempts lose value	Telephone OR 0.62; 13.0% at first call vs 5.5% at six or more; OR 0.93 per doubling	Call mobiles first; stop after three attempts	Calls per subscription
4. Mass calling converts poorly	Phase 1: 3,954 calls per month at 4.8%; ρ = −0.67 between volume and rate	Call the model's top 20% to 30% instead of everyone	Share of subscribers reached
5. Economic effects do not carry forward	Phase-3 AUC 0.60-0.70 vs 0.80 on random test; `nr.employed` outside training range	Refit before each campaign; check AUC monthly	AUC on the latest month

6. Ethics, privacy and Indigenous data sovereignty

6.1 Consent and privacy

The data links personal and credit details to call outcomes, so APP 7 applies: direct marketing needs customer consent or a reasonable expectation, and each campaign needs a simple opt-out (OAIC, 2019). The dataset has no consent or opt-out record, so the company cannot show that customers agreed. From 10 December 2026, privacy policies must also disclose programs that substantially and directly support decisions that significantly affect individuals (Privacy and Other Legislation Amendment Act 2024). A model that decides who receives offers may meet this test.

6.2 Fairness, value and risk

Both models put 98.6% of test customers aged 65+ on the top-20% call list, so a group that makes up 1.8% of customers would receive 8.7% of calls. A high response rate does not show that the plan suits them, and older customers can be more exposed to pressure on the phone. The model also reduces harm: calling the top 30% avoids 70% of calls, mostly to people who would say no. Default status adds little accuracy (importance 0.002), so I recommend removing it.

6.3 Indigenous data sovereignty

The data has no field identifying Aboriginal and Torres Strait Islander customers, so nobody can check how the model treats them. About 43% of 1,545 remote First Nations communities and homelands have no mobile service (RMIT University, 2023). A model that favours cellular contact would rank them low without anyone noticing.

Adding an identifier would not fix this on its own. The Maiam nayri Wingara principles give Aboriginal and Torres Strait Islander people control of the data ecosystem and the right to data that support self-determination (Maiam nayri Wingara & AIGI, 2018). The CARE principles add collective benefit and authority to control (Carroll et al., 2020). Such data should be collected only with free, prior and informed consent, governed with Aboriginal community-controlled organisations, and reported back to communities (Table 7). I would also avoid building proxies for Indigeneity from location or contact type, because group-level inferences can harm a community even when individuals are anonymous.

Table 7. Ethical risks and recommended controls for business stakeholders and end users

Risk	Evidence	Control	Owner
No consent or opt-out record	Dataset has no consent field	Record consent and opt-outs; remove opted-out customers before scoring	Privacy officer
Automated decision not disclosed	Model ranks who receives offers	Describe the model in the privacy policy before 10 December 2026; keep model documentation	Legal and analytics
Over-targeting older customers	98.6% of customers aged 65+ selected	Cap the share of calls per age band; cooling-off period; track complaints and cancellations by age	Marketing manager
Unneeded financial data	`default_unk` importance 0.002	Remove from the deployed model	Analytics team
First Nations customers invisible	No identifier; 43% of remote communities have no mobile service	Do not infer Indigeneity; co-design any data collection with community-controlled organisations under Maiam nayri Wingara and CARE; report results back	Executive sponsor

7. Limitations and conclusion

The variables match public bank telemarketing data collected in Portugal from 2008 to 2010 (Moro et al., 2014), so results may not transfer to current Australian customers. The models predict who says yes on a call; they cannot show that the call caused the subscription.

The most responsive segments are previous subscribers, customers under 25 or over 65, and customers reached by mobile (RQ1; Table 5). A logistic model ranks customers before a call with test AUC 0.80, but accuracy falls to 0.67 in a new phase (RQ2). The next step is a one-month trial that calls a model-ranked list and an equal random list, then compares subscription and complaint rates by age band (RQ3).

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Appendix A. Tests of association with subscription

Table A1. Chi-square tests (Cramér's V) and Mann-Whitney U tests (rank-biserial r), full cleaned data (n = 41,168)

Variable	Test	Statistic	df	p-value	Effect size
poutcome	Chi-square	4217.7	2	<0.001	0.32
month	Chi-square	3103.7	9	<0.001	0.275
job	Chi-square	960.0	11	<0.001	0.153
contact	Chi-square	860.8	1	<0.001	0.145
default	Chi-square	406.2	2	<0.001	0.099
education	Chi-square	192.4	7	<0.001	0.068
marital	Chi-square	123.2	3	<0.001	0.055
day_of_week	Chi-square	26.0	4	<0.001	0.025
housing	Chi-square	5.8	2	0.056	0.012
loan	Chi-square	1.1	2	0.584	0.005
age	Mann-Whitney U	82,840,420	-	0.015	0.022
campaign	Mann-Whitney U	75,347,760	-	<0.001	0.11
euribor3m	Mann-Whitney U	43,474,104	-	<0.001	0.487
nr.employed	Mann-Whitney U	42,537,496	-	<0.001	0.498
cons.price.idx	Mann-Whitney U	65,941,150	-	<0.001	0.221
cons.conf.idx	Mann-Whitney U	91,039,638	-	<0.001	0.075

Appendix B. Logistic regression model selection

Table B1. Nested logistic models fitted by maximum likelihood on the training set (n = 32,934)

Model	Parameters	Log-likelihood	AIC	BIC	McFadden R^2
M1 customer profile	27	-10,953.6	21,961.2	22,188.1	0.055
M2 + contact history	31	-10,040.2	20,142.5	20,402.9	0.134
M3 + economy	34	-9,374.2	18,816.4	19,102.1	0.191
M4 + timing	47	-9,195.2	18,484.4	18,879.3	0.207
M4 using euribor3m	47	-9,205.2	18,504.4	18,899.3	0.206
M5 + contact x previous outcome	49	-9,195.1	18,488.3	18,900.0	0.207
M6 + contact x age group	52	-9,191.6	18,487.1	18,924.1	0.207
M4 without age	42	-9,202.0	18,488.0	18,840.9	0.206
M4 with linear age	43	-9,202.0	18,489.9	18,851.2	0.206
M4 with age + age^2	44	-9,195.3	18,478.5	18,848.2	0.207

Table B2. Likelihood-ratio tests

Comparison	LR χ^2	df	p-value
M1 customer profile → M2 + contact history	1826.8	4	< 0.001
M2 + contact history → M3 + economy	1332.1	3	< 0.001

Comparison	LR χ^2	df	p-value
M3 + economy → M4 + timing	358.0	13	< 0.001
M4 + timing → M5 + contact x previous outcome	0.1	2	0.940
M4 + timing → M6 + contact x age group	7.3	5	0.201
M4 without age → M4 + timing	13.6	5	0.018
M4 with linear age → M4 with age + age^2	13.4	1	< 0.001

Appendix C. Decision tree segments

Table C1. Leaves of a depth-3 decision tree (minimum 300 customers per leaf), training set

Segment rule	Customers	Share of customers	Subscription rate	Share of subscribers
nr.employed <= 5,087.7 and poutcome = success and cons.price.idx > 93.2	523	1.6%	76.7%	10.8%
nr.employed <= 5,087.7 and poutcome = success and cons.price.idx <= 93.2	380	1.2%	64.2%	6.6%
nr.employed <= 5,087.7 and poutcome != success and contact = cellular	2,602	7.9%	38.5%	27.0%
nr.employed > 5,087.7 and cons.conf.idx <= -46.6 and poutcome != failure	1,673	5.1%	23.6%	10.6%
nr.employed <= 5,087.7 and poutcome != success and contact != cellular	482	1.5%	22.6%	2.9%
nr.employed > 5,087.7 and cons.conf.idx <= -46.6 and poutcome = failure	496	1.5%	11.1%	1.5%
nr.employed > 5,087.7 and cons.conf.idx > -46.6 and cons.price.idx <= 94.0	17,096	51.9%	6.8%	31.3%
nr.employed > 5,087.7 and cons.conf.idx > -46.6 and cons.price.idx > 94.0	9,682	29.4%	3.6%	9.3%

Appendix D. Bayesian segment estimates

Table D1. Posterior subscription rates (prior Beta(2.25, 17.75)); last column uses the uniform prior Beta(1, 1)

Segment	n	Observed	Posterior mean	95% credible interval	P(rate > 11.3%)	Uniform prior 95% CrI
Age: 17-24	1,067	24.0%	23.8%	21.3% to 26.3%	1.000	0.215 to 0.266
Age: 25-34	13,684	12.2%	12.2%	11.6% to 12.7%	1.000	0.116 to 0.127
Age: 35-44	13,492	8.6%	8.7%	8.2% to 9.1%	0.000	0.082 to 0.091
Age: 45-54	8,701	8.6%	8.6%	8.1% to 9.2%	0.000	0.081 to 0.093
Age: 55-64	3,563	13.6%	13.5%	12.4% to 14.7%	1.000	0.125 to 0.147
Age: 65+	661	47.4%	46.3%	42.6% to 50.0%	1.000	0.436 to 0.512
Job: blue-collar	9,252	6.9%	6.9%	6.4% to 7.4%	0.000	0.064 to 0.074
Job: retired	1,716	25.2%	25.1%	23.1% to 27.1%	1.000	0.232 to 0.273
Job: student	875	31.4%	31.0%	28.0% to 34.0%	1.000	0.284 to 0.346

Segment	n	Observed	Posterior mean	95% credible interval	P(rate > 11.3%)	Uniform prior 95% CrI
Education: illiterate	18	22.2%	16.5%	6.6% to 29.6%	0.802	0.091 to 0.456
Education: university.degree	12,162	13.7%	13.7%	13.1% to 14.3%	1.000	0.131 to 0.143
Previous outcome: failure	4,250	14.2%	14.2%	13.2% to 15.3%	1.000	0.132 to 0.153
Previous outcome: nonexistent	35,547	8.8%	8.8%	8.5% to 9.1%	0.000	0.085 to 0.091
Previous outcome: success	1,371	65.1%	64.3%	61.8% to 66.8%	1.000	0.625 to 0.675
Contact: cellular	26,131	14.7%	14.7%	14.3% to 15.2%	1.000	0.143 to 0.152
Contact: telephone	15,037	5.2%	5.2%	4.9% to 5.6%	0.000	0.049 to 0.056

Appendix E. Full odds ratio table (model M4)

Table E1. Odds ratios with 95% confidence intervals (reference levels: age 35-44, admin., divorced, university.degree, cellular, nonexistent, May, Friday)

Term	OR	95% CI	p-value
Age 17-24	1.302	1.034 to 1.639	0.025
Age 25-34	1.134	1.023 to 1.258	0.017
Age 45-54	1.094	0.972 to 1.231	0.135
Age 55-64	1.179	1.004 to 1.386	0.045
Age 65+	1.480	1.112 to 1.968	0.007
Job: blue-collar	0.881	0.758 to 1.024	0.100
Job: entrepreneur	1.007	0.798 to 1.269	0.955
Job: housemaid	0.926	0.701 to 1.225	0.592
Job: management	0.978	0.830 to 1.151	0.788
Job: retired	1.170	0.928 to 1.476	0.185
Job: self-employed	0.934	0.747 to 1.169	0.552
Job: services	0.886	0.753 to 1.043	0.146
Job: student	1.146	0.907 to 1.448	0.255
Job: technician	0.996	0.870 to 1.139	0.950
Job: unemployed	0.871	0.676 to 1.121	0.284
Job: unknown	1.128	0.729 to 1.746	0.590
Marital: married	1.084	0.950 to 1.237	0.233
Marital: single	1.107	0.953 to 1.286	0.185
Marital: unknown	1.373	0.635 to 2.967	0.420
Education: basic.4y	0.898	0.753 to 1.071	0.230
Education: basic.6y	1.056	0.856 to 1.302	0.610
Education: basic.9y	0.852	0.729 to 0.995	0.044
Education: high.school	0.936	0.834 to 1.050	0.257
Education: professional.course	0.939	0.813 to 1.084	0.387

Term	OR	95% CI	p-value
Education: unknown	1.058	0.867 to 1.291	0.581
Contact: telephone	0.618	0.541 to 0.704	< 0.001
Previous outcome: failure	0.631	0.559 to 0.711	< 0.001
Previous outcome: success	3.684	3.166 to 4.287	< 0.001
Month: apr	1.997	1.730 to 2.305	< 0.001
Month: aug	1.650	1.397 to 1.948	< 0.001
Month: dec	2.296	1.585 to 3.324	< 0.001
Month: jul	2.319	1.990 to 2.701	< 0.001
Month: jun	2.478	2.142 to 2.869	< 0.001
Month: mar	4.315	3.445 to 5.407	< 0.001
Month: nov	1.389	1.182 to 1.632	< 0.001
Month: oct	1.592	1.274 to 1.990	< 0.001
Month: sep	1.154	0.901 to 1.477	0.256
Day: mon	0.824	0.727 to 0.933	0.002
Day: thu	1.017	0.901 to 1.148	0.787
Day: tue	0.999	0.881 to 1.133	0.989
Day: wed	1.123	0.993 to 1.271	0.065
default_unk (default not 'no')	0.788	0.695 to 0.893	< 0.001
log2(campaign)	0.930	0.891 to 0.971	< 0.001
nr.employed (per 1 unit)	0.990	0.989 to 0.990	< 0.001
cons.price.idx	0.982	0.904 to 1.067	0.667
cons.conf.idx	1.020	1.009 to 1.030	< 0.001

Appendix F. Code

All analysis code is in the notebook QuangTrungNguyen_Assessment3_Code_01112026.ipynb, submitted with this report. It runs from top to bottom with data.csv in the same folder (random seed 42) and recreates every table and figure. Sections: 1 setup and loading; 2 data checks and cleaning; 3 tests of association; 4 train/test split; 5 logistic regression (MLE, model selection, assumptions, odds ratios); 6 non-parametric models and tuning; 7 model comparison (CV, test, bootstrap, calibration, phase-3 test); 8 permutation importance, partial dependence and tree segments; 9 Bayesian estimation (segment rates; logistic regression by MCMC with diagnostics and prior sensitivity); 10 business insight calculations; 11 fairness check; 12 hypothesis summary.

Appendix G. Bayesian logistic regression compared with maximum likelihood

Table G1. Odds ratios from maximum likelihood (95% CI) and from the Bayesian model (posterior median and 95% credible interval), with MCMC diagnostics

Term	MLE OR	95% CI	Bayes OR	95% CrI	R-hat	ESS
Previous outcome: success	3.684	3.166 to 4.287	3.686	3.170 to 4.326	1.000	1,356
Previous outcome: failure	0.631	0.559 to 0.711	0.627	0.556 to 0.709	1.001	1,498
Contact: telephone	0.618	0.541 to 0.704	0.614	0.536 to 0.700	1.001	1,483
Age 17-24	1.302	1.034 to 1.639	1.301	1.032 to 1.637	1.002	1,619
Age 25-34	1.134	1.023 to 1.258	1.133	1.022 to 1.250	1.000	1,663
Age 45-54	1.094	0.972 to 1.231	1.092	0.968 to 1.226	1.001	1,537
Age 55-64	1.179	1.004 to 1.386	1.177	0.994 to 1.380	1.000	1,424
Age 65+	1.480	1.112 to 1.968	1.477	1.114 to 1.974	1.000	1,682
Job: student	1.146	0.907 to 1.448	1.152	0.908 to 1.442	1.002	1,596
Job: retired	1.170	0.928 to 1.476	1.169	0.926 to 1.475	1.001	1,512
Job: blue-collar	0.881	0.758 to 1.024	0.882	0.757 to 1.034	1.001	1,418
default_unk (default not 'no')	0.788	0.695 to 0.893	0.790	0.695 to 0.893	1.002	1,538
log2(campaign)	0.930	0.891 to 0.971	0.929	0.890 to 0.970	1.001	1,574
nr.employed (per 1 unit)	0.990	0.989 to 0.990	0.990	0.989 to 0.990	1.000	1,500
Month: mar	4.315	3.445 to 5.407	4.310	3.403 to 5.448	1.001	1,574
Day: mon	0.824	0.727 to 0.933	0.823	0.729 to 0.937	1.001	1,618

Note. Normal(0, 2.5²) priors; 4 chains, 4,400 retained draws. All 46 terms are in tables/bayes_logit_vs_mle.csv. With the tighter Normal(0, 0.5²) prior the largest shift in a posterior mean is 1.1 standard errors (median 0.24); for the terms in Table 6 it is 0.6 or less.